

[Secure Exit](#)SIMATS
ENGINEERING

SIMATS Online Programming Lab

JAVA PROGRAMMING

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Array

Q2

Numbers

Q3

Control

Q4

Strings

4 / 4 saved

Write a program for matrix multiplication?

Sample Input:

Mat1 =

1 2

5 3

Mat2 =

2 3

4 1

Sample Output:

Mat Sum =

10 5

22 18

Your Code

```
1 import java.util.Scanner;
2 class matrix{
3     public static void main(String args[]){
4         int r1,c1,r2,c2;
5         Scanner sc = new Scanner(System.in);
6         r1 = sc.nextInt();
7         c1 = sc.nextInt();
8         r2 = sc.nextInt();
9         c2 = sc.nextInt();
10        int a[][]=new int[r1][c1];
11        int b[][]=new int[r2][c2];
12        int result[][]=new int[r1][c2];
13        if(c1!=r2){
14            System.out.println("matrix multi
15            return;
```

```
16         }
17
18     for(int i=0;i<r1;i++){
19         for(int j=0;j<c1;j++){
20             a[i][j]=sc.nextInt();
21         }
22     }
23     for(int i=0;i<r2;i++){
24         for(int j=0;j<c2;j++){
25             b[i][j]=sc.nextInt();
26         }
27     }
28     for(int i=0;i<r1;i++){
29         for(int j=0;j<c2;j++){
30             for(int k=0;k<c1;k++){
31                 result[i][j]+=a[i][k]*b[k][j];
32             }
33         }
34     }
35     for(int i=0;i<r1;i++){
36         for(int j=0;j<c2;j++){
37             System.out.println(result[i][j]);
38         }
39         System.out.println("");
40     }
41     sc.close();
42 }
43
```

Program Input (stdin)

```
1
2
5
3
```

Output

```
10
5

22
18
```

 Run

 Save

 Submit

Next →

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